

CSE260: Digital Logic Design

Assignment 2

Summer 2025

1. Why are integrated circuits (ICs) preferred over the classical gate-level method?
2. Differentiate between **gate-level design** and **block-level design** methods.
3. What are the four common and useful MSI circuits, and what are their functions?
4. How does a 2-to-4 decoder work? Explain with its truth table.
5. How can a **demultiplexer** be realized using a decoder with an enable input?
6. Derive the output expression of a 4-to-1 multiplexer.
7. What is the difference between a normal encoder and a **priority encoder**? Give an example.
8. How can you implement a full adder using a 3×8 decoder?
9. Explain how larger multiplexers can be constructed using smaller ones, build an 8-to-1 mux from 4-to-1 mux
10. Design a 18 person voting system using encoder(s) and decoder(s) and parallel adder(s). You can not use full adders. Meaning, for the inputs, you **MUST** use encoder(s) and decoder(s) as required. Make sure your circuit is optimized.